Internship Project-4:-

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Load dataset
df = pd.read_csv("Dataset for Data Analytics - Sheet1.csv")

# Display data
print(df.head())

# Bar Chart
plt.figure(figsize=(8,5))
df['Category'].value_counts().plot(kind='bar')
plt.title('Category Distribution')
plt.xlabel('Category')
plt.ylabel('Count')
plt.savefig('bar_chart.png')
plt.show()

# Histogram
plt.figure(figsize=(8,5))
plt.hist(df['Sales'], bins=10)
plt.title('Sales Distribution')
plt.xlabel('Sales')
plt.ylabel('Frequency')
plt.savefig('histogram.png')
plt.show()
```

```
# Scatter Plot
plt.figure(figsize=(8,5))
plt.scatter(df['Sales'], df['Profit'])
plt.title('Sales vs Profit')
plt.xlabel('Sales')
plt.ylabel('Profit')
plt.savefig('scatter_plot.png')
plt.show()

# Heatmap
plt.figure(figsize=(8,5))
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap='coolwarm')
plt.title('Correlation Heatmap')
plt.savefig('heatmap.png')
plt.show()
```

Expected Output for each:-

1. Bar Chart Expected Output

Output

A vertical bar chart showing category-wise comparison.

Example

If your dataset has categories like:

Technology

Furniture

Office Supplies

The chart will display bars with different heights based on count or sales.

What You Should Observe

- Which category is highest
- Which category is lowest
- Comparison between categories

Expected Insight Example

- Technology category has the highest number of entries.
- Furniture category appears less frequently.

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2. Line Chart Expected Output

Output

A line graph showing trend or progression over time/index.

Example

If using sales data:

100

120

150

130

180

The line moves upward and downward showing trends.

What You Should Observe

- Increasing trend
- Decreasing trend
- Sudden spikes or drops

Expected Insight Example

- Sales increased overall.
- Some fluctuations are visible.

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3. Pie Chart Expected Output

Output

A circular chart divided into slices representing percentages.

Example

Technology → 40% Furniture → 35% Office Supplies → 25%

What You Should Observe

- Largest contribution
- Smallest contribution

- Percentage distribution

Expected Insight Example

- Technology contributes the largest share.
- Office Supplies contributes the least.

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4. Histogram Expected Output

Output

A histogram with bars representing frequency ranges.

Example

If sales values range from 100–1000:

The histogram shows how many values fall within each range.

What You Should Observe

- Most common value range
- Distribution shape
- Outliers

Expected Insight Example

- Most sales fall in the medium range.
- Very high sales values are fewer.

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5. Scatter Plot Expected Output

Output

Dots scattered across the graph showing relationship between two columns.

Example

Sales on X-axis and Profit on Y-axis.

What You Should Observe

- Positive relationship
- Negative relationship
- Outliers

Expected Insight Example

- Higher sales generally lead to higher profit.
- Some points show losses despite high sales.

6. Heatmap Expected Output

Output

A colored matrix showing correlation values.

Example

Correlation values:

1.0

0.8

-0.3